

MSDM5004 Spring 2026

Part II Assignment 1

Due: 30 Apr 23:59

1. Detecting periodicity

The file sunspots.txt contains the observed number of sunspots on the Sun for each month since January 1749. The file contains two columns of numbers, the first representing the month and the second being the sunspot number.

- (a) Write a program that reads the data in the file and makes a graph of sunspots as a function of time. You should see that the number of sunspots has fluctuated on a regular cycle for as long as observations have been recorded. Make an estimate of the length of the cycle in months.
- (b) Modify your program to calculate the Fourier transform of the sunspot data and then make a graph of the magnitude squared $|c_k|^2$ of the Fourier coefficients as a function of k --- also called the *power spectrum* of the sunspot signal. You should see that there is a noticeable peak in the power spectrum at a nonzero value of k . The appearance of this peak tells us that there is one frequency in the Fourier series that has a higher amplitude than the others around it, which corresponds to the periodic wave you can see in the original data.
- (c) Find the approximate value of k to which the peak corresponds. What is the period with this value of k ? You should find that the period corresponds roughly to the length of the cycle that you estimated in part (a).

This kind of Fourier analysis is a sensitive method for detecting periodicity in signals. Even in cases where it is not clear to the eye that there is a periodic component to a signal, it may still be possible to find one using a Fourier transform

2. Fourier filtering and smoothing

The file dow.txt contains the daily closing value for each business day from late 2006 until the end of 2010 of the Dow Jones Industrial Average, colloquially called “the Dow”, which is a measure of average prices on the US stock market.

Write a program to do the following:

- (a) Read in the data from dow.txt and plot them on a graph.
- (b) Calculate the coefficients of the discrete Fourier transform of the data.
- (c) Now set all but the first 10% of the elements of this array to zero (i.e., set the last 90% to zero but keep the values of the first 10%).

- (d) Calculate the inverse Fourier transform of the resulting array, zeros and all, and plot it on the same graph as the original data. You may need to vary the colors of the two curves to make sure they both show up on the graph. Comment on what you see. What is happening when you set the Fourier coefficients to zero?
- (e) Modify your program so that it sets all but the first 2% of the coefficients to zero and run it again.
The particular time period of dow.txt was special in one sense: the value of the Dow at the end of the period was almost the same as at the start, so the function was, roughly speaking, periodic. Another file called dow2.txt also contains data on the Dow but for a different time period, from 2004 until 2008. Over this period the value changed considerably from a starting level around 9000 to a final level around 14000.
- (f) Write a program similar to the one for part (e), in which you read the data in the file dow2.txt and plot it on a graph. Then smooth the data by calculating its Fourier transform, setting all but the first 2% of the coefficients to zero, and inverting the transform again, plotting the result on the same graph as the original data. As in part (e), you should see that the data are smoothed, but now there will be an additional artifact. At the beginning and end of the plot you should see large deviations away from the true smoothed function. These occur because the function is required to be periodic --- its last value must be the same as its first --- so it needs to deviate substantially from the correct value to make the two ends of the function meet. In some situations (including this one) this behavior is unsatisfactory. If we want to use the Fourier transform for smoothing, we would certainly prefer that it does not introduce artifacts of this kind.
- (g) Modify your program to repeat the same analysis using discrete cosine transforms. Again discard all but the first 2% of the coefficients, invert the transform, and plot the result. You should see a significant improvement, with less distortion of the function at the ends of the interval. This occurs because the cosine transform does not force the value of the function to be the same at both ends.

It is because of the artifacts introduced by the strict periodicity of the DFT that the cosine transform is favored for many technological applications, such as audio compression. The artifacts can degrade the sound quality of compressed audio and the cosine transform generally gives better results.

The cosine transform is not wholly free of artifacts itself however. It's true it does not force the function to be periodic, but it does force the gradient to be zero at the ends of the interval (which the ordinary Fourier transform does not). You may be able to see this in your calculations for part (g) above. Look closely at the smoothed function and you should see that its slope is flat at the beginning and end of the

interval. The distortion of the function introduced is less than the distortion in part (f), but it's there all the same.

To reduce this effect, audio compression schemes often use overlapped cosine transforms, in which transforms are performed on overlapping blocks of samples, so that the portions at the ends of blocks, where the worst artifacts lie, need not be used.

3. The file `blur.txt` contains a 1024×1024 square grid of values representing brightness on a black-and-white blurry photo. It is known that the camera used to take the photo has a Gaussian point spread function (PSF)

$$f(x, y) = \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right),$$

with width $\sigma = 25$.

In this question, you will deblur it by circular deconvolution.

- (a) Write a program that reads the grid of values into a two dimensional array of real numbers and then draws the values on the screen of the computer as a density plot. You should see the photo appear. If you get something wrong it might be upside-down. Work with the details of your program until you get it appearing correctly. (Hint: The picture has the sky, which is bright, at the top and the ground, which is dark, at the bottom.)
- (b) Write another program that creates an array, of the same size as the photo, containing a grid of samples drawn from the Gaussian above. Make a density plot of these values on the screen too, so that you get a visualization of your point spread function. Remember that the point spread function is approximated by its periodic summation and is therefore periodic (along both axes). Since the Gaussian is centered at the origin, this means there should be bright patches in each of the four corners of your picture, something like this:



(c) Combine your two programs and add Fourier transforms to make a program that does the following:

- (i) Reads in the blurred photo
- (ii) Calculates the point spread function
- (iii) Fourier transforms both
- (iv) Divides one by the other
- (v) Performs an inverse transform to get the deblurred photo
- (vi) Displays the deblurred photo on the screen

When you are done, you should be able to make out the scene in the photo, although probably it will still not be perfectly sharp.

Hint: One thing you'll need to deal with is what happens when the Fourier transform of the point spread function is zero, or close to zero. In that case if you divide by it you'll get an error (because you can't divide by zero) or just a very large number (because you're dividing by something small). A workable compromise is that if a value in the Fourier transform of the point spread function is smaller than a certain amount ε you don't divide by it --- just leave that coefficient alone. The value of ε is not very critical but a reasonable value is 10^{-3} .

4. Solve the Poisson Equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = \rho(x, y)$$

in $0 \leq x \leq 1, 0 \leq y \leq 2$ by the spectral method taught in class, with $\Delta \leq 0.01$.

The boundary condition is

$$\left\{ \begin{array}{l} \phi(x=0, 0 < y < 2) = 0 \\ \phi(x=1, 0 < y < 2) = 0 \\ \phi(0 \leq x \leq 1, y=0) = 0 \\ \phi(0 \leq x \leq 1, y=2) = \sqrt{1-x^2} \end{array} \right.$$

The source density is

$$\rho(x, y) = \begin{cases} 16\pi & 0.25 \leq x < 0.5, 1.25 \leq y < 1.5 \\ 0 & \text{Otherwise} \end{cases}$$

Show your answer by plotting $\phi(x, y)$.